GOLPHIN ANNO S

251901049

# COMPUTER NETWORKS

| Ex.No.1 | **Study of various Network commands used in Linux and Windows: Hands-on practice of various network commands.** |
|---|---|
| Date: | |

**AIM:** - To Study of various Network commands used in Linux and Windows

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| **1** | `arp -a` | `arp -a` | Displays the ARP cache, showing the IP addresses and corresponding MAC addresses of devices on the local network. | `arp -a` | `Interface: 192.168.1.10 --- 0x7 Internet Address Physical Address Type 192.168.1.1 00-1a-2b-3c-4d-5e dynamic` |
| **2** | `hostname` | `hostname` | Displays the hostname (computer name) of the local system. | `hostname` | `DESKTOP-PC01` |
| **3** | `ipconfig /all` | `ipconfig /all` | Displays complete TCP/IP configuration, including IP address, subnet mask, gateway, DNS, DHCP, and MAC address. | `ipconfig /all` | `IPv4 Address : 192.168.1.10 Default Gateway : 192.168.1.1 DNS Server : 8.8.8.8 DHCP Enabled : Yes` |
| **4** | `nbtstat -a` | `nbtstat -a <hostname>` | Displays NetBIOS name table of a remote computer for troubleshooting NetBIOS over TCP/IP. | `nbtstat -a DESKTOP-PC01` | `NetBIOS Remote Machine Name Table Name Type Status DESKTOP-PC01 <00> UNIQUE Registered` |
| **5** | `netstat` | `netstat` | Displays active TCP/IP | `netstat` | `Proto Local Address Foreign Address State TCP` |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| | | | connections, listening ports, routing tables, and interface statistics. | | 192.168.1.10:49762 142.250.190.78:443 ESTABLISHED |
| **6** | netstat -r | netstat -r | Displays the IP routing table of the local computer. | netstat -r | IPv4 Route Table Network Destination Netmask Gateway Interface |
| **7** | nslookup | nslookup <domain> | Queries the DNS server to obtain the IP address or DNS records of a domain. | nslookup www.google.com | Server: dns.google Address: 8.8.8.8 Name: www.google.com Address: 142.250.190.68 |
| **8** | pathping | pathping <hostname/IP> | Combines the features of ping and tracert to trace the route and measure packet loss at each hop. | pathping www.google.com | Tracing route to www.google.com Computing statistics... Hop 1 192.168.1.1 Hop 2 10.0.0.1 Packet Loss: 0% |
| **9** | ping | ping <hostname/IP> | Tests connectivity between the local computer and another host using ICMP Echo Request and Reply messages. | ping www.facebook.com | Pinging www.facebook.com [157.240.22.35] with 32 bytes of data: Reply from 157.240.22.35: bytes=32 time=25ms TTL=54 |
| **10** | route | route print | Displays or modifies the IP routing table. It is commonly used to view or configure static routes. | route print | IPv4 Route Table Active Routes: Network Destination Netmask Gateway Interface |

## Some important Linux networking commands

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| 1 | ip | ip <OPTIONS> <OBJECT> <COMMAND> | Displays and manages IP addresses, routing tables, and network interfaces. | ip address show | 2: eth0: <UP,BROADCAST,RUNNING> inet 192.168.1.100/24 brd 192.168.1.255 scope global eth0 |
| 2 | ip address add | ip address add <IP>/<Prefix> dev <Interface> | Assigns an IP address to a network interface. | ip address add 192.168.1.254/24 dev eth0 | Assigned 192.168.1.254/24 to eth0 successfully. |
| 3 | ip address del | ip address del <IP>/<Prefix> dev <Interface> | Removes an IP address from a network interface. | ip address del 192.168.1.254/24 dev eth0 | Deleted 192.168.1.254/24 from eth0. |
| 4 | ip link set up | ip link set <Interface> up | Brings the network interface online. | ip link set eth0 up | Interface eth0 is UP. |
| 5 | ip link set down | ip link set <Interface> down | Brings the network interface offline. | ip link set eth0 down | Interface eth0 is DOWN. |
| 6 | ip link set promisc on | ip link set <Interface> promisc on | Enables promiscuous mode on the interface. | ip link set eth0 promisc on | Promiscuous mode enabled on eth0. |
| 7 | ip route add default | ip route add default via <Gateway> dev <Interface> | Adds the default gateway for all outgoing traffic. | ip route add default via 192.168.1.254 dev eth0 | Default route added successfully. |
| 8 | ip route add | ip route add <Network> via <Gateway> | Adds a static route through a gateway. | ip route add 192.168.1.0/24 via 192.168.1.254 | Route added successfully. |
| 9 | ip route add (Direct Route) | ip route add <Network> dev <Interface> | Adds a direct route through a network interface. | ip route add 192.168.1.0/24 dev eth0 | Route added via eth0. |
| 10 | ip route delete | ip route delete <Network> via <Gateway> | Deletes a static route. | ip route delete 192.168.1.0/24 | Route deleted successfully. |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| | | | | via 192.168.1.254 | |
| 11 | ip route get | ip route get <Destination IP> | Displays the route used to reach a destination. | ip route get 10.10.1.4 | 10.10.1.4 via 192.168.1.254 dev eth0 src 192.168.1.100 |
| 12 | ifconfig | ifconfig | Displays and configures network interface parameters (deprecated in modern Linux). | ifconfig | eth0: flags=4163<UP,BROADCAST,RUNNING> inet 192.168.1.100 netmask 255.255.255.0 |
| 13 | mtr | mtr <hostname/IP> | Combines Ping and Traceroute to analyze network paths. | mtr google.com | HOST: localhost Loss% Snt Last Avg Best Wrst StDev |
| 14 | mtr -g | mtr -g <hostname> | Displays numeric IP addresses instead of hostnames. | mtr -g google.com | 192.168.1.1 → 142.250.190.78 |
| 15 | mtr -b | mtr -b <hostname> | Displays both hostnames and IP addresses. | mtr -b google.com | router.local (192.168.1.1) google.com (142.250.190.78) |
| 16 | mtr -c | mtr -c <count> <hostname> | Sends a specified number of probes. | mtr -c 10 google.com | 10 probes completed successfully. |
| 17 | tcpdump -D | tcpdump -D | Lists all available network interfaces. | tcpdump -D | 1.eth0 2.lo 3.wlan0 |
| 18 | tcpdump | tcpdump -i <Interface> | Captures packets on a network interface. | tcpdump -i eth0 | Listening on eth0... |
| 19 | tcpdump -c | tcpdump -i <Interface> -c <count> | Captures a specified | tcpdump -i eth0 -c 10 | 10 packets captured. |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| | | | number of packets. | | |
| 20 | tcpdump host | `tcpdump -i <Interface> host <IP>` | Captures traffic to and from a specific host. | `tcpdump -i eth0 host 8.8.8.8` | `Captured packets to/from 8.8.8.8.` |
| 21 | tcpdump src host | `tcpdump -i <Interface> src host <IP>` | Captures packets originating from a specific host. | `tcpdump -i eth0 src host 8.8.8.8` | `Source packets from 8.8.8.8 displayed.` |
| 22 | tcpdump dst host | `tcpdump -i <Interface> dst host <IP>` | Captures packets destined for a specific host. | `tcpdump -i eth0 dst host 8.8.8.8` | `Destination packets to 8.8.8.8 displayed.` |
| 23 | tcpdump net | `tcpdump -i <Interface> net <Network>/<Prefix>` | Captures traffic from a specified network. | `tcpdump -i eth0 net 10.1.0.0/24` | `Packets from network 10.1.0.0/24 displayed.` |
| 24 | tcpdump port | `tcpdump -i <Interface> port <Port>` | Captures traffic on a specific port. | `tcpdump -i eth0 port 53` | `DNS packets captured.` |
| 25 | tcpdump host and port | `tcpdump -i <Interface> host <IP> and port <Port>` | Captures traffic for a specific host and port. | `tcpdump -i eth0 host 8.8.8.8 and port 53` | `DNS traffic with host 8.8.8.8 displayed.` |
| 26 | tcpdump port not | `tcpdump -i <Interface> port not <Port1> and not <Port2>` | Captures all traffic except the specified ports. | `tcpdump -i eth0 port not 53 and not 25` | `Traffic excluding ports 53 and 25 captured.` |
| 27 | ping | `ping <hostname/IP>` | Tests network connectivity using ICMP Echo Requests and Echo Replies. | `ping google.com` | `64 bytes from google.com: icmp_seq=1 ttl=117 time=10.5 ms` |
| 28 | ping -c | `ping -c <count> <hostname>` | Sends a specified number of | `ping -c 10 google.com` | `10 packets transmitted, 10 received, 0% packet loss.` |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| | | | ICMP Echo Requests. | | |

**Configuring an Ethernet connection by using nmcli:**

| Step No. | Task | Command | Expected Result |
|---|---|---|---|
| **1** | List all Network Manager connection profiles | `nmcli connection show` | Displays all available network connection profiles with their **Name, UUID, Type, and Device**. |
| **2** | Create a new Ethernet connection profile<br><br>It is specifically an **`nmcli` connection management command**.<br><br>• **`nmcli`** → NetworkManager Command Line Interface<br>• **`connection`** → Manages network connection profiles.<br>• **`add`** → Creates a new connection profile. | `nmcli connection add con-name Internal-LAN ifname enp1s0 type ethernet` | Creates a new Ethernet connection profile named **Internal-LAN**. |
| **3** | Rename an existing connection profile (Optional) | `nmcli connection modify "Wired connection 1" connection.id Internal-LAN` | Renames the existing connection profile to **Internal-LAN**. |
| **4** | Display the current settings of the connection profile | `nmcli connection show "Internal-LAN"` | Displays interface name, autoconnect status, IPv4 and IPv6 configuration details. |
| **5** | Configure IPv4 using DHCP | `nmcli connection modify "Internal-LAN" ipv4.method auto` | Configures the interface to obtain the IPv4 address |

| Step No. | Task | Command | Expected Result |
|---|---|---|---|
| | | | automatically using DHCP. |
| **6** | Configure a static IPv4 address | `nmcli connection modify "Internal-LAN" ipv4.method manual ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com` | Assigns a static IPv4 address, gateway, DNS server, and DNS search domain. |
| **7** | Configure IPv6 using SLAAC | `nmcli connection modify "Internal-LAN" ipv6.method auto` | Enables automatic IPv6 configuration using Stateless Address Autoconfiguration (SLAAC). |
| **8** | Configure a static IPv6 address | `nmcli connection modify "Internal-LAN" ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::ffee ipv6.dns 2001:db8:1::ffbb ipv6.dns-search example.com` | Assigns a static IPv6 address, gateway, DNS server, and search domain. |
| **9** | Activate the connection profile | `nmcli connection up Internal-LAN` | Activates the **Internal-LAN** network connection profile. |
| **10** | Verify the IP address configuration | `ip address show enp1s0` | Displays the assigned IPv4 and IPv6 addresses of the **enp1s0** network interface. |
| **11** | Verify the IPv4 default gateway | `ip route show default` | Displays the configured IPv4 default gateway. |
| **12** | Verify the IPv6 default gateway | `ip -6 route show default` | Displays the configured IPv6 default gateway. |
| **13** | Verify the DNS configuration | `cat /etc/resolv.conf` | Displays the configured DNS servers and DNS search domain. |
| **14** | Test network connectivity | `ping <host-name-or-IP-address>` | Sends ICMP Echo Requests to verify connectivity with the specified host. |

**Sample Verification Outputs**

| Verification Command | Sample Output |
|---|---|
| ip address show enp1s0 | inet 192.0.2.1/24<br>inet6 2001:db8:1::fffe/64 |
| ip route show default | default via 192.0.2.254 dev enp1s0 proto static |
| ip -6 route show default | default via 2001:db8:1::ffee dev enp1s0 proto static |
| cat /etc/resolv.conf | search example.com<br>nameserver 192.0.2.200<br>nameserver 2001:db8:1::ffbb |
| ping google.com | 64 bytes from google.com: icmp_seq=1 ttl=117 time=11.2 ms |

**Output:**

```
student@fedora:~$ arp -a
_gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3
student@fedora:~127$ hostname
fedora
```

```
student@fedora:~127$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 fedora:bootpc           _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     22939
unix  3      [ ]         STREAM     CONNECTED     22757    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     22589    /run/user/1000/at-spi/bus_0
unix  2      [ ]         DGRAM                    18681    /run/systemd/home/notify
unix  3      [ ]         STREAM     CONNECTED     19513    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     23704
unix  3      [ ]         STREAM     CONNECTED     23002
unix  3      [ ]         STREAM     CONNECTED     21482    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     21176    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     19398
unix  3      [ ]         DGRAM      CONNECTED     17375
unix  2      [ ]         DGRAM      CONNECTED     16660
unix  3      [ ]         STREAM     CONNECTED     19667
unix  3      [ ]         STREAM     CONNECTED     21249    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     19763    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     22770    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     23282    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     23497    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     22269    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     23113    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     22914    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     22635    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     21191
unix  3      [ ]         STREAM     CONNECTED     23738
unix  3      [ ]         STREAM     CONNECTED     21192
unix  3      [ ]         STREAM     CONNECTED     22680
unix  3      [ ]         STREAM     CONNECTED     22610    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     22922
unix  3      [ ]         STREAM     CONNECTED     21199    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     23540    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     22761    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     23011
unix  3      [ ]         STREAM     CONNECTED     22887    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     21391    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     23725    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     22489
```

```
student@fedora:~$ netstat -r
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
default         _gateway        0.0.0.0         UG        0 0          0 enp0s3
```

```
student@fedora:~2$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 380  bytes 52722 (51.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 468  bytes 45472 (44.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 39  bytes 3622 (3.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 39  bytes 3622 (3.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
student@fedora:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.67.46
Name:   google.com
Address: 2404:6800:4007:805::200e
```

```
student@fedora:~127$ ping facebook.com
PING facebook.com (57.144.56.1) 56(84) bytes of data.
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=1 ttl=255 time=14.2 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=2 ttl=255 time=21.2 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=3 ttl=255 time=8.87 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=4 ttl=255 time=16.2 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=5 ttl=255 time=13.4 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=6 ttl=255 time=19.7 ms
^C
--- facebook.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5192ms
rtt min/avg/max/mdev = 8.867/15.598/21.155/4.092 ms
```

```
student@fedora:~$ route print
Usage: route [-nNvee] [-FC] [<AF>]           List kernel routing tables
       route [-v] [-FC] {add|del|flush} ...  Modify routing table for AF.

       route {-h|--help} [<AF>]              Detailed usage syntax for specified AF
       route {-V|--version}                  Display version/author and exit.

        -v, --verbose            be verbose
        -n, --numeric            don't resolve names
        -e, --extend             display other/more information
        -F, --fib                display Forwarding Information Base (default)
        -C, --cache              display routing cache instead of FIB

  <AF>=Use -4, -6, '-A <af>' or '--<af>'; default: inet
  List of possible address families (which support routing):
    inet (DARPA Internet) inet6 (IPv6) ax25 (AMPR AX.25)
    netrom (AMPR NET/ROM) ipx (Novell IPX) ddp (Appletalk DDP)
    x25 (CCITT X.25)
```

```
student@fedora:~3$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 83951sec preferred_lft 83951sec
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac/64 scope global dynamic noprefixroute
       valid_lft 85914sec preferred_lft 13914sec
    inet6 fe80::55f0:9771:cb55:7bf3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

```
student@fedora:~$ mtr -g google.com
mtr: invalid option -- 'g'

Usage:
 mtr [options] hostname

 -F, --filename FILE        read hostname(s) from a file
 -4                         use IPv4 only
 -6                         use IPv6 only
 -u, --udp                  use UDP instead of ICMP echo
 -T, --tcp                  use TCP instead of ICMP echo
 -I, --interface NAME       use named network interface
 -a, --address ADDRESS      bind the outgoing socket to ADDRESS
 -f, --first-ttl NUMBER     set what TTL to start
 -m, --max-ttl NUMBER       maximum number of hops
 -U, --max-unknown NUMBER   maximum unknown host
 -P, --port PORT            target port number for TCP, SCTP, or UDP
 -L, --localport LOCALPORT  source port number for UDP
 -s, --psize PACKETSIZE     set the packet size used for probing
 -B, --bitpattern NUMBER    set bit pattern to use in payload
 -i, --interval SECONDS     ICMP echo request interval
 -G, --gracetime SECONDS    number of seconds to wait for responses
 -Q, --tos NUMBER           type of service field in IP header
 -e, --mpls                 display information from ICMP extensions
 -Z, --timeout SECONDS      seconds to keep probe sockets open
 -M, --mark MARK            mark each sent packet
 -r, --report               output using report mode
 -w, --report-wide          output wide report
 -c, --report-cycles COUNT  set the number of pings sent
 -j, --json                 output json
 -x, --xml                  output xml
 -C, --csv                  output comma separated values
 -l, --raw                  output raw format
 -p, --split                split output
 -t, --curses               use curses terminal interface
     --displaymode MODE     select initial display mode
 -n, --no-dns               do not resolve host names
 -b, --show-ips             show IP numbers and host names
 -o, --order FIELDS         select output fields
 -y, --ipinfo NUMBER        select IP information in output
 -z, --aslookup             display AS number
 -h, --help                 display this help and exit
 -v, --version              output version information and exit
```

**RESULT:**

**The various network commands in Linux and Windows were studied and executed successfully.**